

SHOBDON AF, United Kingdom

WMO: 035200

Lat: 52.2431N

Lon: 2.8858W

Elev: 99

StdP: 100.14

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-4.3	-2.7	-5.8	2.3	-1.3	-4.5	2.6	-0.5	13.4	9.6	12.2	9.6	0.9	70	0.568

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.2	25.7	19.2	23.8	18.1	22.1	17.1	20.6	23.1	18.9	22.3	17.7	21.1	3.2	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.0	14.8	21.3	17.7	12.8	19.9	16.5	11.8	19.1	59.9	23.0	53.9	22.3	50.3	21.3	29.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.4	9.0	7.8	-7.9	28.5	2.9	2.0	-9.9	29.9	-11.6	31.1	-13.3	32.2	-15.4	33.6
			-7.8	23.2	3.0	2.3	-10.0	24.8	-11.8	26.2	-13.5	27.5	-15.7	29.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.8	4.5	4.9	6.3	8.5	11.3	14.2	16.1	15.8	13.4	10.5	7.1	4.7
	DBStd	5.09	3.28	3.09	2.87	2.70	2.79	2.50	2.31	2.37	2.33	2.90	3.20	3.85
	HDD10.0	807	173	143	117	61	17	1	0	0	2	29	97	168
	HDD18.3	3129	430	375	372	296	217	126	76	84	147	243	338	423
	CDD10.0	736	1	1	4	14	59	127	189	181	106	44	8	3
	CDD18.3	17	0	0	0	0	0	2	7	7	1	0	0	0
	CDH23.3	209	0	0	0	0	6	36	103	58	6	1	0	0
	CDH26.7	27	0	0	0	0	0	4	14	9	0	0	0	0
Wind	WSAvg	3.1	3.6	3.7	3.6	3.1	3.1	2.8	2.7	2.6	2.6	2.9	3.0	3.3
Precipitation	PrecAvg	827	89	65	67	57	60	62	47	65	71	78	81	91
	PrecMax	1047	157	142	148	131	168	256	114	183	199	218	181	188
	PrecMin	563	10	3	3	4	5	6	6	6	14	11	27	23
	PrecStd	109	42	38	30	31	33	49	26	36	43	44	38	44
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.8	13.4	16.3	20.8	24.2	26.9	28.4	28.0	24.1	19.1	15.1	13.6
		MCWB	11.0	11.3	11.2	14.5	17.4	19.1	19.4	21.0	19.4	15.6	13.4	11.9
	2%	DB	11.7	11.9	13.8	17.5	21.4	23.9	26.1	24.9	21.4	17.3	14.1	12.6
		MCWB	10.2	10.1	10.4	12.8	16.3	17.7	18.9	19.8	16.6	14.7	12.6	11.1
	5%	DB	10.6	10.9	12.4	15.4	19.1	21.8	24.1	22.6	19.7	16.2	13.2	11.6
		MCWB	9.1	9.1	9.8	11.6	14.6	16.5	18.3	17.9	15.6	14.0	11.6	10.2
	10%	DB	9.6	9.9	11.3	13.7	17.0	19.9	22.1	20.9	18.4	15.2	12.1	10.5
		MCWB	8.1	8.1	9.0	10.3	13.1	15.5	17.2	16.9	14.9	13.2	10.6	9.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.6	11.7	12.6	15.5	19.4	20.9	23.8	24.8	20.5	16.7	13.9	12.2
		MCDB	12.5	12.7	13.8	18.9	21.2	24.2	24.1	25.1	22.6	18.6	14.8	13.1
	2%	WB	10.4	10.3	11.2	13.4	16.9	18.4	20.3	20.6	17.6	15.4	12.8	11.4
		MCDB	11.5	11.7	13.1	16.5	20.9	22.4	23.8	22.3	20.1	16.7	13.9	12.4
	5%	WB	9.3	9.4	10.2	12.1	15.2	17.1	18.6	18.7	16.5	14.4	11.9	10.4
		MCDB	10.4	10.7	12.1	14.7	18.1	20.7	23.0	21.4	18.8	15.8	13.0	11.5
	10%	WB	8.3	8.3	9.2	10.9	13.9	16.0	17.5	17.5	15.5	13.4	10.8	9.2
		MCDB	9.5	9.7	11.0	13.2	16.3	19.3	21.3	20.3	17.7	14.8	12.0	10.4
Mean Daily Temperature Range	5% DB	MDBR	5.8	6.2	7.6	8.9	9.4	9.7	10.2	9.8	9.3	7.6	6.5	5.8
		MCDBR	6.3	6.8	9.5	12.1	12.8	13.5	14.5	12.5	11.6	8.2	6.6	6.0
		MCWBR	5.7	5.5	6.8	7.8	8.1	7.9	8.3	7.8	7.6	5.9	5.6	5.5
	5% WB	MCDBR	6.3	6.7	8.6	10.8	11.3	11.7	12.3	11.0	10.2	7.6	6.4	6.0
		MCWBR	5.9	5.8	6.8	7.8	8.2	8.0	8.5	8.6	7.7	6.0	5.7	5.6
Clear-Sky Solar Irradiance	taub	0.318	0.334	0.366	0.389	0.396	0.389	0.386	0.372	0.361	0.345	0.320	0.308	
	taud	2.350	2.382	2.293	2.241	2.259	2.306	2.317	2.375	2.407	2.427	2.401	2.322	
	Ebn at Noon	674	779	819	843	856	864	859	853	819	755	669	612	
	Edh at Noon	62	79	104	123	127	123	119	107	92	75	58	53	
All-Sky Solar Radiation	RadAvg	0.74	1.40	2.52	3.87	4.79	5.14	4.79	3.93	2.98	1.68	0.88	0.57	
	RadStd	0.07	0.16	0.31	0.45	0.27	0.56	0.51	0.32	0.23	0.15	0.09	0.07	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (80 neighbors)	N/A	N/A	N/A	N/A	-1.30	-1.54	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page